

Denis Meleshkin

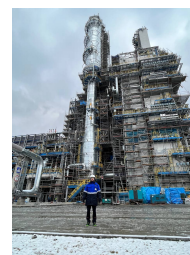
✉ melesh.denis@gmail.com

🏠 Padova, Italy

🌐 LinkedIn profile

🌐 Portfolio

Curriculum vitae



Working Experience

- 2022 – 2023 📌 **Jr. Design Engineer.** PJSC "Cryogenmash".
The main area of work was the calculation and modelling of cryogenic systems in Aspen HYSYS software, and rating of heat exchangers.
- 2021 – 2022 📌 **Purchasing Manager.** LLC STATMED.
The main responsibilities included: market analysis, interaction with partners, organization of international purchases of various volumes.

Internships

- 2022 📌 **Production practice at the Amur Gas Processing Plant.**
During the internship the operation of the natural gas purification, separation and liquefaction line, as well as the production of liquid helium, was studied.
- 2021 📌 **Technological practice at the NPO Nauka enterprise.**
During the internship a technological process for manufacturing a pressure valve part was developed.

Education

- 2023 – 2025 📌 **M.Sc. Energy Engineering.**
University of Padua, Industrial Engineering department.
Thesis title: *Numerical investigation on the use of Twisted Tape turbulators for water single-phase forced convection.*
Final grade 100/110.
- 2018 – 2022 📌 **B.Sc. Refrigeration, Cryogenic Equipment And Life Support Systems.**
Bauman Moscow State Technical University, Power Engineering department
Thesis title: *Helium cryogenic cooling system with a capacity of 2 kW per 100-150 K.*
Diploma with honour.

Publications

Journal Articles

- **D. Meleshkin**, S. Pochatkov, and I. Svetlov, "Vibrodiagnostics method for determining damage to composite materials," *Grand Mechanic*, no. 5, pp. 337–342, 2022. 🔗 URL: <https://www.elibrary.ru/item.asp?id=48445134>.

Conference Proceedings

- D. Ershov, A. Mironov, **D. Meleshkin**, S. Pochatkov, and I. Svetlov, "Methods of numerical analysis and vibration diagnostics for assessing deformations in mechanical systems," in *Students' scientific spring (2022)*, Moscow, Russia, 2022, pp. 22–24. 🔗 URL: <https://www.elibrary.ru/item.asp?id=49480593>.

Skills

Engineering	• Design and analysis of energy systems and devices, modelling and simulation; • Planning and carrying out an experiment, processing the results; • Entropy and exergy analysis; • Life Cycle Assessment; • Developing of the design documentation.
Software	• Calculations: MATLAB, PTC Mathcad, MS Excel, Julia; • Modelling and simulations: Simulink and Simscape, ANSYS, OpenFOAM, Aspen HYSYS; • Design and documentation: Autodesk Inventor, Autodesk AutoCAD, Solid-Edge, CFTurbo; • Reporting and publishing: L ^A T _E X, MS Word, MS PowerPoint, MS Visio, GIT.
Language	Strong reading, writing and speaking competencies for English, proved by IELTS.
Misc.	Academic research, working in groups, management skills. Leadership experience: during the studying for the Bachelor's degree, I was elected head of the study group, held this honorary position until graduation and was awarded a special payment for it.